

PROJECT REPORT

Auto Ticket Classification using Flow Designer

Automated IT ticket categorisation and email notification on the ServiceNow platform

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1. INTRODUCTION

1.1 Project Overview

IT support teams receive tickets that contain only a short description of the problem. Someone then has to read each ticket and set its Category and Subcategory by hand before it can be routed. This project automates that step on the ServiceNow platform.

The solution has four parts:

- A custom table, Incident WorkFlow (u_incident_workflow), stores every ticket and generates a unique number with the prefix INC.
- Category and Subcategory are choice fields with a dependency, so only valid Subcategory options appear for the selected Category.
- A Flow Designer flow, Auto Classify School IT Tickets, runs when a ticket is created with an empty Category. It reads the Short Description, matches keywords, and fills in Category and Subcategory.
- A Send Email action then confirms the ticket to the caller.

The whole solution is built with configuration only (no custom scripting) and is packaged in an update set so it can be moved to another instance.

1.2 Purpose

The purpose of this project is to remove manual ticket triage for common IT issues. The system is designed to:

- Classify IT tickets automatically from the issue description.
- Assign Category and Subcategory without manual intervention.
- Keep Category and Subcategory consistent through dependent choice logic.
- Send an automatic email to the caller when a ticket is created.
- Store ticket information in a structured and standard format.
- Be easy to maintain and extend, for example by adding one more Else If branch for a new issue type.

2. IDEATION PHASE

2.1 Problem Statement

Support staff and callers lose time on manual ticket classification, and inconsistent categories lead to wrong routing and delayed responses. The table below states the problem from the caller's and the support team's point of view.

Who	I am	I am trying to	But	Because	Which makes me feel
Caller (student / staff)	A person facing an IT problem such as Wi-Fi or a projector not working	Report the issue and know that it has been received	I get no confirmation and do not know if it was categorised correctly	Tickets are triaged by hand and confirmations are not automatic	Uncertain and frustrated
IT support agent	A member of the IT support team	Pick up correctly categorised tickets quickly	I must read and classify each ticket myself	Callers write free text and categories are set manually	Overloaded and slowed down

2.2 Empathy Map Canvas

The empathy map below is for the main user, the caller who raises an IT ticket.

Says	Thinks
"My Wi-Fi is not working." "Did anyone get my request?" "Please fix the projector before class."	Will my issue reach the right person? Why do I have to choose so many options? I hope I get a reply soon.
Does	Feels
Opens the ticket form and types a short description. Waits for an update or checks the status again. Calls or visits the IT desk when there is no response.	Anxious when there is no confirmation. Relieved when an acknowledgement arrives. Frustrated by delays and repeated follow-ups.
Pains	Gains
No acknowledgement after submitting. Wrong or missing category delays the fix. Having to explain the issue more than once.	Instant email confirmation. Ticket categorised correctly without effort. Faster routing and resolution.

2.3 Brainstorming

The team listed possible ways to remove manual classification and compared them.

Idea	Description	Decision
Manual triage by an agent	An agent reads each ticket and sets the category.	Rejected: slow and inconsistent.
Drop-down only for the caller	Caller selects Category and Subcategory on the form.	Kept as a dependent choice, but not relied on alone, because callers often skip or misselect.
Keyword rules in Flow Designer	Match words in the Short Description and set Category and Subcategory.	Selected: simple, no code, easy to maintain.
Machine-learning classifier	Train a model on past tickets.	Postponed: needs historical data and a separate setup. Listed under Future Scope.
Email confirmation	Notify the caller when a ticket is created.	Selected: added as the last action of the flow.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Caller action	System response	Caller feeling
1. Report	Opens the Incident WorkFlow form, picks the caller and types a short description.	Assigns a unique INC number on save.	Neutral
2. Classification	Does nothing further.	The flow detects an empty Category, matches keywords and sets Category and Subcategory.	Relieved: no extra steps
3. Confirmation	Checks email.	Sends a confirmation to the caller's email address.	Reassured
4. Assignment	Waits for the support team.	Ticket is ready for Assigned group and Assigned to to be set.	Hopeful
5. Resolution	Receives the fix.	State is updated through New, In progress, On hold, Resolved, Closed.	Satisfied

3.2 Solution Requirement

Functional requirements

ID	Requirement	How it is met
FR-1	Create a ticket with caller and description	Custom table u_incident_workflow and its form
FR-2	Generate a unique ticket number	Auto Number with prefix INC, starting at 500, 5 digits
FR-3	Provide Category and Subcategory choices	Choice fields with Network, Hardware, Access, Performance and Wi-Fi, Projector, Forgot Password, Slow Computer
FR-4	Subcategory depends on Category	Dependent field set on Subcategory with dependent values
FR-5	Classify tickets automatically	Flow Designer flow with If / Else If conditions on Short description
FR-6	Notify the caller by email	Send Email action to Caller's email address
FR-7	Move the solution between instances	Update set exported as XML

Non-functional requirements

Type	Requirement
Usability	No extra steps for the caller; classification happens in the background.

Type	Requirement
Maintainability	Each issue type is one Else If branch, so changes are made in one place without code.
Scalability	New categories, keywords and subcategories can be added later.
Data quality	Choice fields and references keep data structured and consistent.
Security	Access controls on the table restrict who can work with ticket data.
Reliability	The flow runs on every record creation where Category is empty.

3.3 Data Flow Diagram

The diagram shows how ticket data moves from the caller to the table, through the flow, and back to the caller as an email.

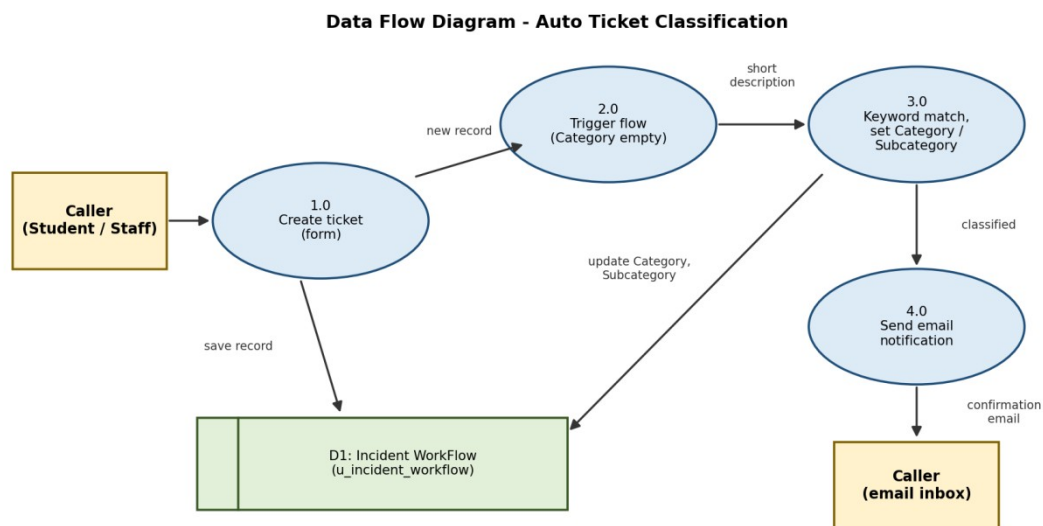


Figure 1: Data flow diagram

3.4 Technology Stack

Component	Technology / Feature
Platform	ServiceNow (Global application scope)
Data storage	Custom table u_incident_workflow with Auto Number
Form design	Form Design and Dictionary configuration, choice lists, dependent fields
Automation	Workflow Studio / Flow Designer: record-created trigger, If / Else If, Update Record
Notification	Send Email action, outbound email logged in System Logs > Emails
Version control / deployment	Local Update Set, exported to XML
Development style	Low-code / no-code, with no custom scripts

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem	Solution fit
Callers must be classified by hand, which is slow	A flow classifies the ticket the moment it is created.
Categories are inconsistent or mismatched	Dependent choices allow only valid Category and Subcategory pairs.
Callers receive no acknowledgement	An email confirmation is sent automatically.
Changes are hard to track and move	All work is captured in one update set and exported as XML.

4.2 Proposed Solution

Aspect	Description
Problem being solved	Manual classification of IT tickets and missing caller confirmation.
Idea	Use keyword rules in Flow Designer to set Category and Subcategory at ticket creation, then email the caller.
Novelty	Fully configuration-based; no scripts and no machine-learning setup are needed, so it is quick to build and easy to hand over.
Value	Less manual effort for support staff, consistent data, and faster routing.
Scalability	More issue types can be supported by adding an Else If branch and matching choices.
Limitation	Matching is by exact keywords, so unusual wording or typos will not be classified.

4.3 Solution Architecture

The solution is organised in layers on the ServiceNow platform.

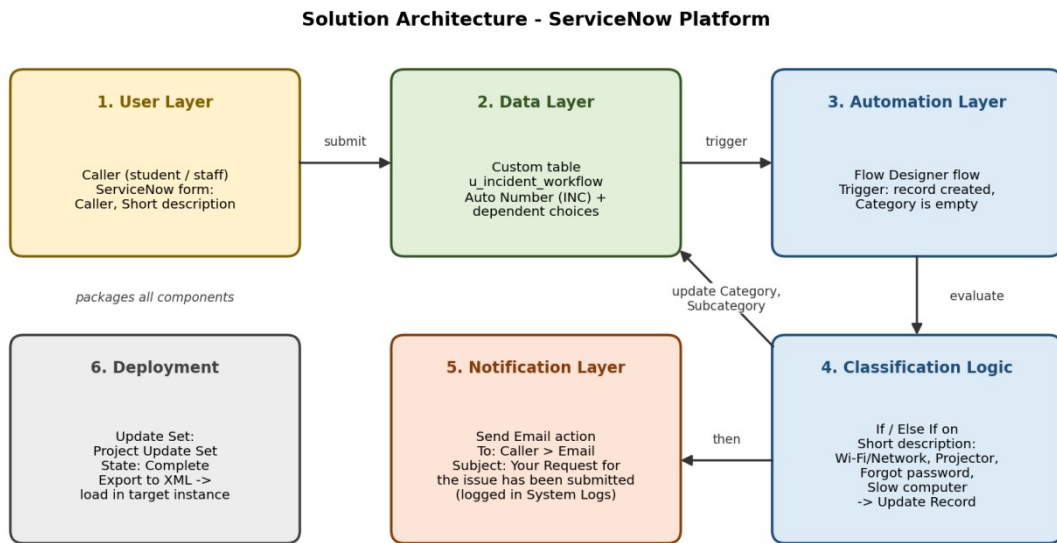


Figure 2: Solution architecture

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The work was carried out in five phases.

Phase	Activities	Output	Dates	Status
1. Planning and setup	Business use case, requirements, create and set the update set	Project Update Set as current set	01-10-2026	Completed
2. Backend development	Create the Incident WorkFlow table, enable Auto Number, create fields and choices	Table u_incident_workflow with fields	02-10-2026	Completed
3. Dependency configuration	Make Subcategory dependent on Category and set dependent values	Filtered Subcategory options	02-10-2026	Completed
4. Flow automation	Build trigger, classification branches and email action; activate flow	Active flow Auto Classify School IT Tickets	02-10-2026	Completed
5. Testing and deployment	Test scenarios, check email log, complete and export the update set	Test evidence and XML file	02-10-2026	Completed

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The solution is rule-based, not a trained machine-learning model. Training accuracy and fine-tuning therefore do not apply, and testing checks that each rule produces the expected result.

Model performance summary

S.No.	Parameter	Values
1	Model Summary	Rule-based automation built with a ServiceNow custom table and a Flow Designer flow. The flow triggers on record creation when Category is empty, checks the Short Description for keywords, then runs Update Record and Send Email.
2	Accuracy	Training Accuracy: not applicable (no training data). Validation Accuracy: 100% on the tested scenarios (2 of 2).
3	Fine Tuning Result (if done)	Not applicable. Rule-level refinements: the trigger condition Category is empty (null) protects manual selections, and dependent values restrict Subcategory options. Validation Accuracy: 100% on the tested scenarios.

Functional test cases

ID	Short description entered	Expected result	Actual result	Status
TC-1	A Wi-Fi / network issue (ticket INC00511)	Category = Network, Subcategory = Wi-Fi, email sent	Network / Wi-Fi; email sent to caller	Pass
TC-2	Projector not turning on (ticket INC00512)	Category = Hardware, Subcategory = Projector, email sent	Hardware / Projector; email sent to caller	Pass
TC-3	Forgot password	Category = Access, Subcategory = Forgot Password	Access / Forgot password; email sent to caller	Pass
TC-4	Slow computer	Category = Performance, Subcategory = Slow Computer	Performance / Slow computer; email sent to caller	Pass

Data validation checks

- Mandatory fields (Caller, Short description) are captured correctly.
- The auto number is generated without duplication.
- Category and Subcategory values are stored accurately.
- Reference fields resolve correctly to the user and group tables.

7. RESULTS

7.1 Output Screenshots

Backend set-up

Update Set
Project Update Set

* Name: Project Update Set
State: In progress
Parent:
Release date:
Install date:
Installed from:
Description:
Application: Global
Created: 2026-02-06 20:55:31
Created by: admin
Merged to:
Update Delete

Update Delete

Related Links
Make This My Current Set
Merge With Another Update Set
Scan Update Set
ISN Update Versions (IO)

Customer Updates Update Set Logs Child Update Sets Install History

Update set = Project Update Set

Created	Type	View	Target name	Updated by	Remote update set	Action
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Figure 3: Project Update Set created in Global scope

Table
New record

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More info](#)

* Label: Incident Workflow
* Name: u_incident_workflow
Extends table:
Application: Global
Create module: ☒
Remote Table: ☐

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Insert a new row...					

Submit Cancel

Related Links

Figure 4: New table Incident Workflow (u_incident_workflow)

Table
Incident Workflow

Columns Controls Application Access

Extensible: ☐
Live feed: ☐

Use auto-numbering to define a sequential identifying code made up of a prefix, a base number and a padding value to ensure a consistent format

Auto-number: ☒
Prefix: INC
Number: 500
Number of digits: 5

Security Rules (ACLs) are required if anyone other than an administrator needs to work with this table. Creating default security rules will grant full access to this table to anyone with the user role you specify.

Create access controls: ☒
* User role: u_incident_workflow_user

Delete Update Delete All Records

Related Links
Form Builder

Figure 5: Auto Number enabled with prefix INC, number 500 and 5 digits

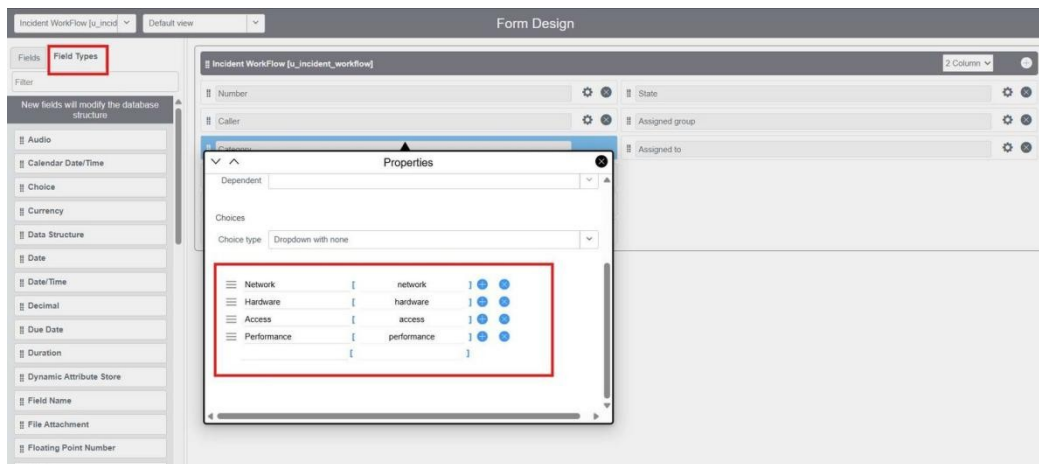


Figure 6: Choices added to the Category field in Form Design

Dependent choices

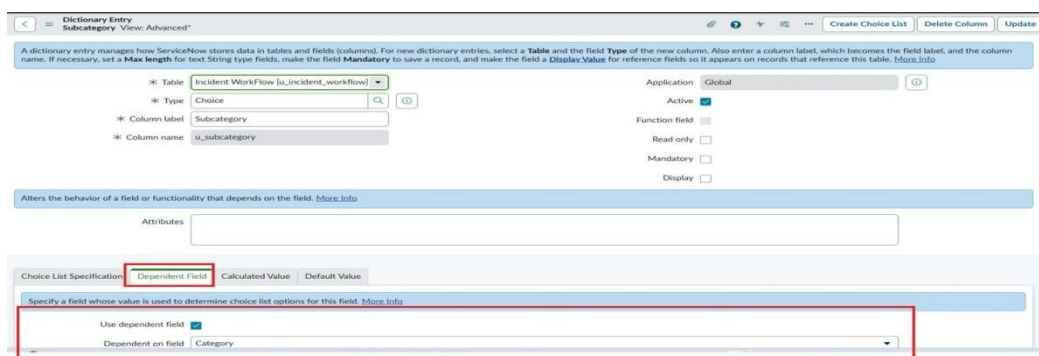


Figure 7: Use dependent field enabled, dependent on Category

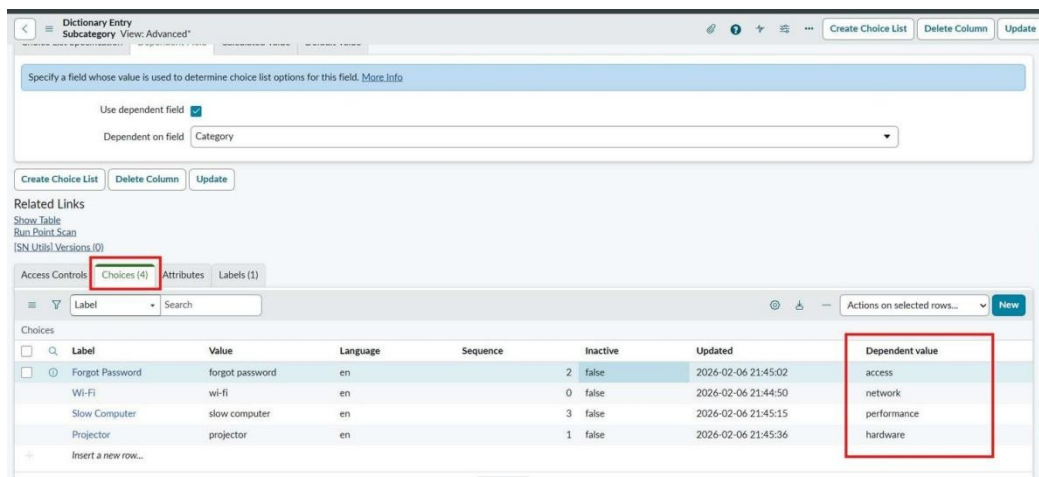


Figure 8: Dependent values set for each Subcategory choice

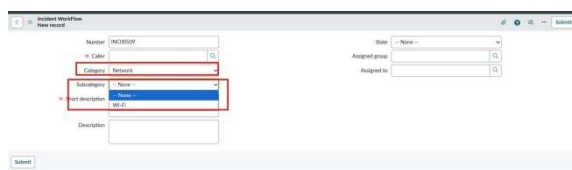


Figure 9a: Network shows Wi-Fi



Figure 9b: Hardware shows Projector

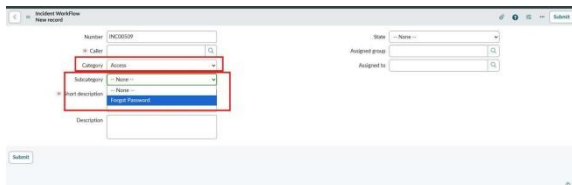


Figure 9c: Access shows Forgot Password

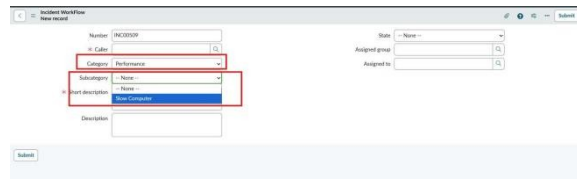


Figure 9d: Performance shows Slow Computer

Flow automation

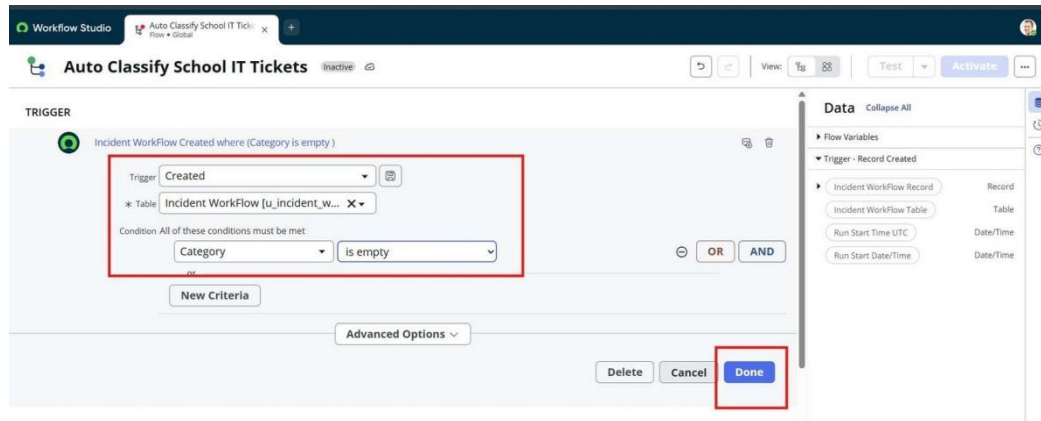


Figure 10: Flow trigger: Record Created on Incident WorkFlow where Category is empty

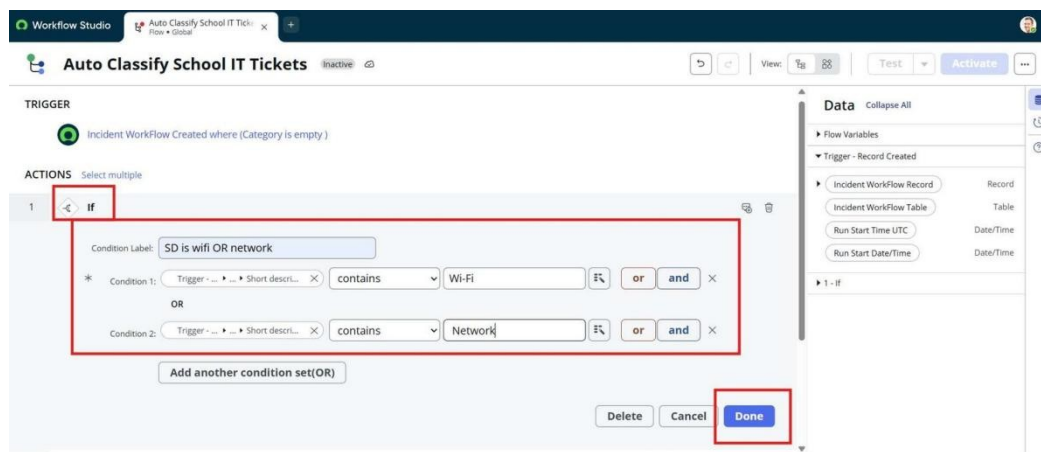


Figure 11: If condition: Short description contains Wi-Fi or Network

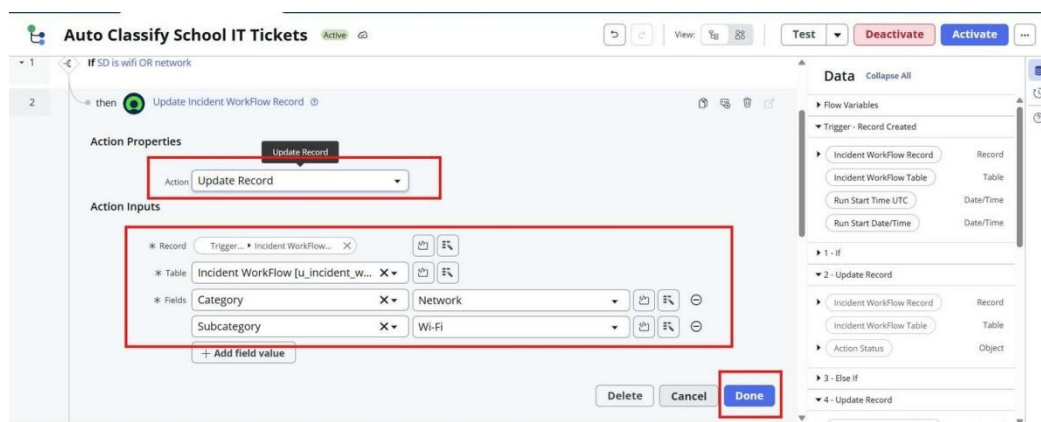


Figure 12: Update Record action: Category = Network, Subcategory = Wi-Fi

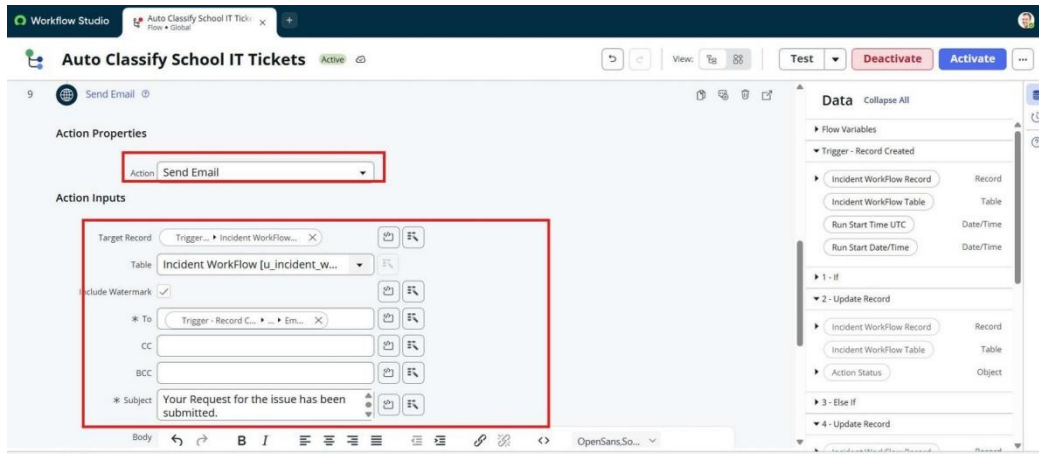


Figure 13: Send Email action to the caller

Testing

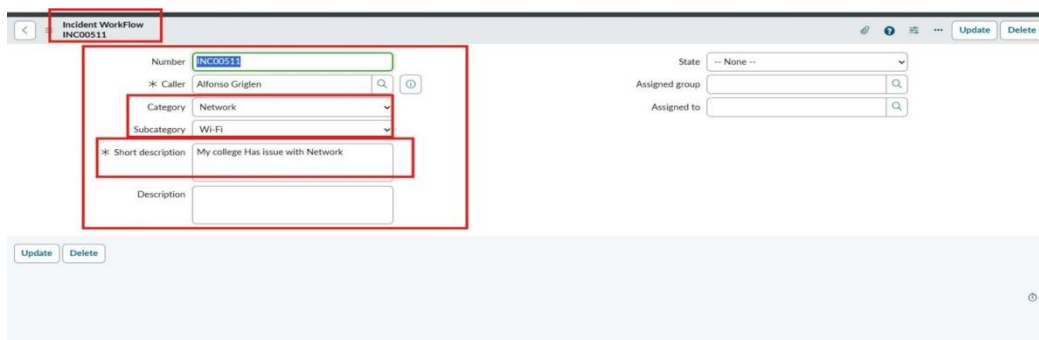


Figure 14: Test scenario 1: ticket INC00511 classified as Network / Wi-Fi



Figure 15: Confirmation emails listed in System Logs > Emails

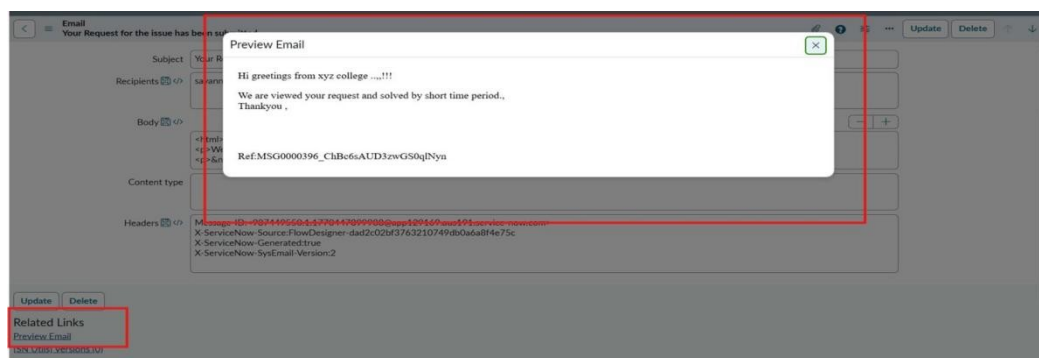


Figure 16: Preview of the confirmation email

Incident Workflow INCD00512

Number: INCD00512

Caller: Sam Sorokin

Category: Hardware

Subcategory: Projector

Short description: Projector not turning on

Description:

State: -- None --

Assigned group:

Assigned to:

Update Delete

Figure 17: Test scenario 2: ticket INCD00512 classified as Hardware / Projector

Deployment

Update Set Project Update Set

Name: Project Update Set

State: Complete

Parent:

Release date:

Install date:

Installed from:

Description:

Application: Global

Created: 2026-02-06 20:55:31

Created by: admin

Merged to:

Update Back Out

Related Links

Export to XML

Scan Update Set

ISN Utilis Versions (0)

Figure 18: Update set state changed to Complete; Export to XML link

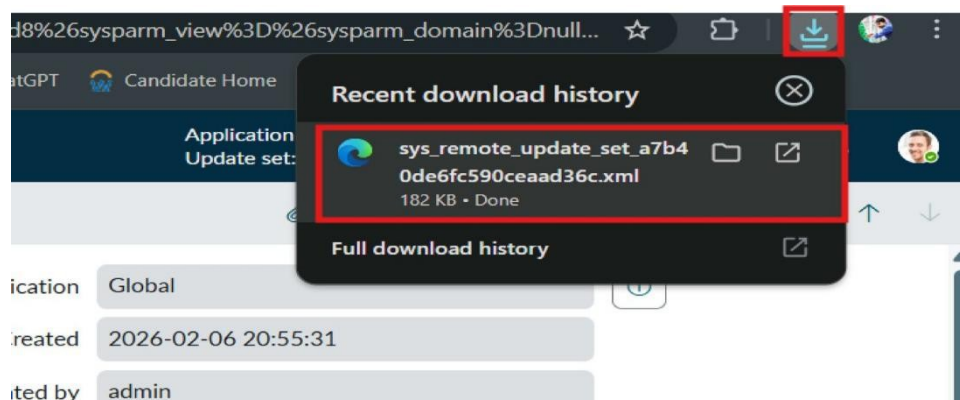


Figure 19: Update set XML file downloaded

8. ADVANTAGES & DISADVANTAGES

Advantages

- Tickets are classified automatically and instantly.
- Category and Subcategory are always a valid pair.
- The caller gets an immediate email confirmation.
- Built without code, so it is easy to understand and maintain.
- New issue types can be added with one more Else If branch.
- The update set makes the solution portable to other instances.

Disadvantages

- Classification depends on exact keywords in the Short description. Different wording or spelling mistakes will not match.
- Only four issue types are covered at present.
- The flow runs only when Category is empty, so it does not reclassify tickets after edits.
- Email delivery depends on the instance's outbound email configuration.

9. CONCLUSION

The project replaces manual triage of common IT tickets with an automated flow on ServiceNow. A custom table stores tickets in a standard format with unique numbers, dependent choices keep Category and Subcategory consistent, and a Flow Designer flow assigns both values from the Short description and emails the caller. Testing with Wi-Fi / network and projector tickets gave the expected Category, Subcategory and email in both cases. The solution is packaged in an update set, so it can be exported and installed on another instance.

10. FUTURE SCOPE

- Add more categories and keywords, such as printer, software installation and email problems.
- Handle spelling variations and synonyms, for example by matching several keywords per issue type.
- Replace keyword rules with a machine-learning classifier trained on past tickets (for example ServiceNow Predictive Intelligence).
- Add automatic assignment of the Assigned group and Assigned to based on the category.
- Set priority and expected resolution time from the category.
- Send status-change emails when the State moves to Resolved or Closed.
- Add dashboards that show ticket counts by category.

11. APPENDIX

Source Code

The solution is configuration-based, so there is no custom source code. The flow logic is summarised below.

```
Flow: Auto Classify School IT Tickets (Global)
Trigger: Record Created on Incident Workflow [u_incident_workflow]
Condition: Category is empty

IF Short description contains "Wi-Fi" OR "Network"
THEN Update Record: Category = Network, Subcategory = Wi-Fi
ELSE IF Short description contains "Projector"
THEN Update Record: Category = Hardware, Subcategory = Projector
ELSE IF Short description contains "Forgot password"
THEN Update Record: Category = Access, Subcategory = Forgot Password
ELSE IF Short description contains "Slow computer"
THEN Update Record: Category = Performance, Subcategory = Slow Computer

Send Email: To = Caller > Email
Subject = Your Request for the issue has been submitted
Body = Ticket confirmation message
```

Item	Details
Dataset link	Not applicable. No dataset is used because the solution is rule-based.
GitHub link	https://github.com/darshinim1985-boop/Auto-Ticket-Classification-using-FLOW-DESIGNER-
Project demo link	https://drive.google.com/file/d/1Jlm42kek4Cq8JtFO0pcNioObZjIJdaFg/view?usp=drivesdk